INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)

SBT-DF205 Mobile and IoT Forensics

Forensic Examination Report: Case 1 Android Forensics (Google Pixel 3, Android 10)

Field	Detail
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Reg Number	2025/FWSD/11468
Course Code	SBT-DF205 (Mobile and IoT Forensics)
Practical Title	Case 1: Android Forensics Lab, Google Pixel 3 (Android 10)
Case Reference	SBT-DF205-C1 | Requesting party (training case): Harbour Operations Ltd
Evidence	Joshua Hickman, Android 10 Pixel 3 file system extraction (Digital Corpora, Non-Cellebrite Extraction), Pixel 3.zip
Examination environment	Kali Linux 2026.2 on VMware Workstation; all times reported in UTC
Report date	27 October 2026

1. Scope and instructions

Harbour Operations Ltd asked for an examination of a company-issued Google Pixel 3 after suspected unauthorised communication and sharing of information. Only the supplied file system extraction was examined: no rooting, resetting or live-account access. The aims were to establish the device users, reconstruct communications, assess media and location evidence, and state whether the records **support, contradict or leave the allegation unresolved**. Original identities and historical dates are used, raw timestamps are kept beside UTC conversions, and no passwords, tokens, keys or verification codes are reported (they were not read or are redacted).

2. Method and evidence handling (Task 1)

Item	Record
Source URL	https://digitalcorpora.s3.amazonaws.com/corpora/mobile/android_10/Non-Cellebrite%20Extraction/Pixel%203.zip (HTTP 200)
Archive and size	`Pixel 3.zip`, 5,247,820,897 bytes; download 2026-10-28 03:09:03Z to 03:51:54Z (receipt time; VM clock EDT, −0400, converted to UTC)
Baseline SHA-256	`ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a` (EV-001; no manifest supplied, so recorded as the baseline)
Integrity rechecks	`sha256sum -c` returned OK at 2026-10-28 04:05:09Z and, after all examination steps, at 2026-10-29 04:19:43Z and 04:22:37Z (Figure 2). The original was made read-only (`chmod a-w`).
Working copy	Extracted to `work/`: 76,403 archive entries, 76,402 items on disk (the difference is the top-level folder entry, `unzip -Z1` check)
Examiner and tools	Mary-Joy Adewole. GNU Wget 1.25.0; SQLite 3.53.4 (upgraded from 3.46.1 before extraction, logged); ExifTool 13.55; UnZip 6.00 (Debian 6.0-31); Python 3.14.7
Register and log	Evidence_Register.csv: 64 entries (EV-040 to EV-042 duplicate EV-037 to EV-039 and are marked as such). Handling_Log.csv: 64 rows including four resolved errors and three corrections.

Databases were opened read-only as copies, always with their `-wal` and `-shm` files, and each copy was verified identical to its source with `cmp`. The effect of the WAL was measured rather than assumed: `bugle_db` returns **36 messages with its WAL and 35 without** (the extra record, `messages._id` 39, is the sent SMS of 2020-02-14 13:09:03 UTC); `wa.db` has **no wa_contacts table without its WAL** (3 rows with it); `msgstore.db`, `gphotos0.db` and `gphotos-1.db` give identical row content with and without the WAL.

```
sha256sum "original/Pixel 3.zip" | tee original/Pixel3.zip.sha256
sha256sum -c original/Pixel3.zip.sha256      # run from the case root -> original/Pixel 3.zip: OK
```

3. Device identity and profiles (Task 2)

Property key	Value	Source (evidence ID)
ro.product.vendor.manufacturer	Google	vendor/build.prop (EV-003)
ro.product.vendor.model / ro.product.vendor.device	Pixel 3 / blueline	vendor/build.prop (EV-003)
ro.build.version.release / ro.build.version.sdk	10 / 29	system/build.prop (EV-002)
ro.build.id	QQ1A.200105.003	system/build.prop (EV-002)
ro.build.version.security_patch	2020-01-01	system/build.prop (EV-002)
ro.system.build.fingerprint	google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-keys	system/build.prop (EV-002); `ro.build.fingerprint` is absent

The Magisk mirror copy of `system/build.prop` is identical (`cmp`), and a `data/magisk_backup_...` folder is present: the handset had been rooted when it was extracted. This is recorded as device state and is not attributed to any user.

Profile (EV)	Name, flags	Created / last logged in (raw ms, then UTC)	Accounts (source)
User 0 (EV-005)	'This Is', flags 19, serial 0	created=0 (none recorded); lastLoggedIn 1581695517124 = 2020-02-14 15:51:57	13 accounts in accounts_ce/de.db (EV-007, EV-008): Google thisisdfir@gmail.com, WhatsApp, Telegram, Signal, Skype, Viber (+19195794674), TextNow, TikTok, imo, Duo, Messenger and others
User 11 (EV-006)	'User 2', flags 16, serial 11	created 1581644358241 = 2020-02-14 01:39:18; lastLoggedIn 1581684601922 = 2020-02-14 12:50:01	Google thisisdfirtwo@gmail.com only (accounts_de.db, EV-009)

Flags follow AOSP `UserInfo`: 19 is primary (1) + admin (2) + initialized (16); 16 is initialized only. `userlist.xml` (EV-004) shows two users, `nextSerialNumber` 12 and `deviceOwnerUserId` -10000, so **no device owner is recorded: "company-issued" comes from the brief, not from the device**. The `thisisdfir@gmail.com` association is corroborated by the Play Store `lastAccount.xml` (EV-013), by the calendar owner (EV-022) and by 90 XML files under the app and system folders; `thisisdfirtwo@gmail.com` has 0 hits elsewhere because `data/system_ce/11` is absent from the extraction. Account entries and logins show what was configured, not who used the handset: profiles can be shared, a login time does not identify a person, and user 11's credential-encrypted data is unavailable.

4. Settings and connectivity (Task 3)

Stored key	Value	File (evidence ID)
adb_enabled (id 221)	1	settings_global.xml, user 0 (EV-010)
development_settings_enabled (id 199)	1	settings_global.xml, user 0 (EV-010)
device_name / bluetooth_name (secure id 65)	Pixel 3 / Pixel 3	settings_global.xml / settings_secure.xml (EV-010, EV-011)
wifi_on / bluetooth_on / airplane_mode_on	1 / 0 / 0	settings_global.xml (EV-010)

Wi-Fi. Both `WifiConfigStore.xml` files (EV-014, EV-015) have an empty `NetworkList` and `wpa_supplicant.conf` (EV-017) has no `network=` blocks, so **no saved network profile is present**. Residual identifiers remain: a `MacMapEntry` keyed by the SSID as displayed, `CcookiesDcastleR5` with `WPA_PSK`, mapped to `ae:f6:c7:a8:a5:c2` (EV-014), and an open-network notifier blacklist entry for `EX6100_NETGEAR_2GEXT` (EV-015). Keys were not read. **Bluetooth** (`bt_config.conf`, EV-016): adapter `7c:d9:5c:ac:a2:ce` and one bonded device `b4:ec:02:73:ff:93`, name "Rouge", DevClass 7936, DevType 1, a link key present (value not read). `TimeCreated` 2020-01-29 15:38:12 has no offset and is **unresolved**. These are saved configuration, not proof that the handset connected to any network or device at a given time.

5. AOSP communications and contacts (Task 4)

Telephony copy (evidence ID)	Size / SHA-256 (first 16)	sms / pdu / part
data/data/com.android.providers.telephony/databases/mmssms.db (EV-018)	118784 B / cd5452f52933252f	24 / 7 / 10
data/user_de/0/com.android.providers.telephony/databases/mmssms.db (EV-019)	118784 B / d0ce9034defe1f2b	0 / 0 / 0 (empty)

The Magisk mirror copies hash identically to these two, and no sidecars exist for the Telephony, contacts, call-log or calendar databases. `sms.date` is in milliseconds and `pdu.date` in seconds (both validated against Google Messages, which stores milliseconds for both). Direction (`type` 1 received, 2 sent) was checked against content. SMS `_id` 1 is absent.

SMS _id	Thread, address	Dir, read	Raw ms = UTC	Content
17	13, 32665	recv, 0	1580757526393 = 2020-02-03 19:18:46	Facebook friend-suggestion notice [link redacted]
20	13, 32665	recv, 0	1581185821854 = 2020-02-08 18:17:01	Facebook notification [link redacted]
21	17, 9195790479	sent, 1	1581273702392 = 2020-02-09 18:41:42	I am on my way to the coffee shop. Do you want anything?
24	13, 32665	recv, 0	1581628344576 = 2020-02-13 21:12:24	Facebook notification [link redacted]
25	17, 919-579-0479	sent, 1	1581685743277 = 2020-02-14 13:09:03	Wow, the messages sync over between accounts.

Conversation (thread 17, +19195790479, 9 messages = 7 MMS + 2 SMS; threads.message_count agrees). 2020-02-01 20:04:53 sent "Here's Signal." (MMS 1); 20:06:04 received "It looks like the messaging is run through the Messages app. Strange." (MMS 2); 20:07:52 sent "It might be an attempt at unified messaging...?" (MMS 3); 20:08:12 received "Maybe?" (MMS 4); 20:09:27 sent "It will be interesting to see where the messages are stored." (MMS 5). 2020-02-09 18:41:42 sent SMS 21; 18:43:08 received "No, thank you" (MMS 6). 2020-02-14 13:09:03

sent SMS 25; 13:10:08 received "That is interesting." (MMS 7). **MMS 1 reconstruction:** `pdu` 1 (`msg_box` 2, `m_type` 128, `pdu.date` 1580587493 s = 2020-02-01 20:04:53) has two `part` rows: part 1 `application/smil` (layout) and part 2 `text/plain` "Here's Signal."; `_data` is empty, so there is no attachment file. Sent MMS carry `+19195794674` as the From address, the handset number (it is also the Viber account name and the WhatsApp `registration_jid`).

Call _id	Number	Type	Raw ms = UTC	Dur (s)	Contact (contacts2.db, EV-020)
11	9195790479	2 outgoing	1581643741051 = 2020-02-14 01:29:01	11	Josh Hickman
12	+19195790479	1 incoming	1581643824061 = 2020-02-14 01:30:24	65	Josh Hickman
13	9195790479	2 outgoing	1581685557104 = 2020-02-14 13:05:57	86	Josh Hickman
14	9195790479	1 incoming	1581691278351 = 2020-02-14 14:41:18	91	Josh Hickman
7, 8	+19195121037	3 missed, 4 voicemail	1581521849939 = 2020-02-12 15:37:29; 15:38:21	0, 3	no match

Both number formats of `+19195790479` normalise to the same contact (`phone_lookup`); this is the one matched number (raw contacts 1, 4, 9 and 10, contact 10). `raw_contacts` holds 10 rows, all "Josh Hickman", one per account source; rows 6 and 7 (Telegram, number `+990841016803084`) are flagged `deleted` = 1. Unmatched: `+15755863247`, `+19195121037`, `+332629243087`, and five calls with a blank number and `presentation` 2 (restricted per `CallLog.Calls`). Call types follow Android `CallLog.Calls` (1 incoming, 2 outgoing, 3 missed, 4 voicemail). **Calendar (EV-022):** `Calendars` 1 (thisisdfir@gmail.com), `CalendarCache` 4, `Colors` 35, three metadata tables 1 each; `Events`, `Attendees`, `CalendarAlerts`, `EventsRawTimes`, `ExtendedProperties`, `Instances` and `Reminders` are **empty**, so the calendar holds no events.

6. Google Messages and Dialer (Task 5)

Bugle _id	sms_message_uri	Telephony row	Time check (ms)	Read: Bugle / Telephony
29	content://sms/17	SMS 17, 32665, received	received_timestamp = date (1580757526393); sent_timestamp = date_sent	0 / 0
32	content://sms/20	SMS 20, 32665, received	1581185821854 = date	0 / 0
33	content://sms/21	SMS 21, 9195790479, sent	1581273702392 = date	1 / 1
37	content://sms/24	SMS 24, 32665, received	1581628344576 = date	0 / 0
39	content://sms/25	SMS 25, 919-579-0479, sent	1581685743277 = date (**WAL-only record**)	1 / 1

Coverage: 24 of 24 SMS and 7 of 7 MMS link to a Google Messages row (31 of 36). Five rows have no Telephony counterpart: `_id` 3, 9, 14 and 19 (protocol 5, status 210, no text, each 1 ms after a message in the same conversation) and `_id` 20 (protocol 3, status 207, 2020-02-01 20:04:32 UTC, 21 s before MMS 1). They appear app-generated; their meaning was not interpreted. `_id` 22 to 24 are absent and `sqlite_sequence` shows 39 against 36 rows, which is consistent with, but does not prove, deleted rows. Google Messages keeps millisecond precision for MMS (`content://mms/1` = 1580587493369) that the Telephony `pdu` table truncates to seconds.

Dialer. `AnnotatedCallLog` (EV-026) holds 12 rows against 14 in the AOSP `calls` table. All 12 match an AOSP call by timestamp (0 ms difference) and `_id`; AOSP `_id` 8 and 10 (type 4 voicemail, 3 s, with a `voicemail_uri`) are absent, and the Dialer has no row missing from AOSP. `voicemail.db` (EV-027) has no message records (`greeting` 0 rows), so it does not explain the gap. Reproduced in SQL (`dialer_reproduce`: aosp_total 14, dialer_total 12, aosp_not_in_dialer 2, dialer_not_in_aosp 0).

```
ATTACH 'extracts/calllog/calllog.db' AS k;
SELECT count(*) FROM k.calls c WHERE NOT EXISTS
  (SELECT 1 FROM AnnotatedCallLog a WHERE abs(a.timestamp - c.date) < 2000);   -- 2
```

7. WhatsApp and TextNow (Task 6)

WhatsApp (2.20.11). `com.whatsapp_preferences_light.xml` (EV-035): `registration_jid` 19195794674, `ph` 9195794674, `push_name` ThisIsDFIR. Participant 19195790479 is "Josh Hickman" (`wa_contacts`, visible only through the WAL). Direction is `key_from_me`; times are milliseconds. Message content is in the legacy `messages` table (19 rows); the new-schema `message` table has 0 rows.

_id	UTC (raw ms)	Dir	Content or media reference	Receipts (UTC)
12	2020-02-08 20:55:14 (1581195314692)	sent	Hi there! I switched over.	server 20:55:15, read 20:55:18
13	20:56:05 (1581195365000)	recv	Awesome!	none
14	20:57:05 (1581195425000)	recv	image/jpeg, 21,890 B; file IMG-20200208-WA0000.jpg (EV-059) shows an Osborne portable computer	none
15	20:57:49 (1581195469433)	sent	The Osbourne!	server 20:57:49, read 20:57:49
16	20:58:12 (1581195492693)	sent	image/jpeg, 87,909 B; file IMG-20200208-WA0001.jpg (EV-060) is a meme: "This baby is fully encrypted / Trust me"	server 20:58:12, read 20:58:13

_id	UTC (raw ms)	Dir	Content or media reference	Receipts (UTC)
17-19	2020-02-09 18:00:14, 18:00:33, 18:05:02	sent	Location shares, types 16, 16, 5: 35.6609133, −78.8798813; 35.6609423, −78.8798167; 35.6609311, −78.8798382	_id 19: device 18:07:24, read 18:15:25

The base64 SHA-256 of each image file equals `messages.media_hash` for rows 14 and 16. **Status validation:** sent `status` 13 occurs on 6 rows, all with server receipts and 6 with read receipts; sent `status` 6 occurs on 3 rows with no receipts; received statuses are 0 (8 rows), -1 (1) and 13 (1). Only these receipt-backed readings are asserted; other WhatsApp versions use different meanings.

TextNow (EV-034, EV-036). Account association: a `com.enflick.android.TextNow.account` entry in `accounts_ce.db` and `userinfo_phone` 9842352054 with status ENABLED in the app preferences. `messages` holds 9 rows, all with `+19842032223`, 2020-02-08 20:00:46 to 20:11:10 UTC (`date` is milliseconds; the seconds reading gives no valid date). Direction 1 is incoming (the system row "Missed call from +19842032223" has direction 1). Sorted by `date`: 20:00:46 in "Hi there!"; 20:01:27 out "Back at it again, I see."; 20:03:06 in "Yep! Have to keep generating data."; 20:05:41 two app rows (types 102 and 201, state 2); 20:08:22 in missed-call notice; 20:09:16 in picture (ColecoVision console, 53,708 B, EV-044); 20:10:25 out picture (Atari 800, 180,093 B, EV-043); 20:11:10 in "I wonder why I can't call you or send you pictures." Picture links are redacted. **Participant correlation:** the TextNow contact list holds `+19195790479` as Josh Hickman, matching contacts2.db, the call log, SMS thread 17 and WhatsApp; but none of the nine messages involves that number. The message counterpart `+19842032223` and the account number `9842352054` have **0 matches** in Google Messages, Telephony, contacts2.db, the call log and the app files searched: a documented non-match.

8. Maps and photographs (Task 7)

Maps entry	Source	Contents	Record type and time
A	gmm_myplaces.sync_item; gmm_sync.sync_item_data corpus 11, client_id 1:0 (EV-045, EV-047)	121 E Tryon Rd, Raleigh NC; 35.734602, −78.636654; same feature ID in both databases	Place record (address and coordinates), not a search or route; timestamp 1554407343604 ms = 2019-04-04 19:49:03 UTC
B	gmm_sync.sync_item_data, client_id 4:0 (EV-047)	M152 Sweet Vista Lane, Holly Springs NC; 35.659596, −78.872848	Place record; no time recorded (0)
C	gmm_sync.sync_item_data, client_id 5:0 (EV-047)	J30 Enterprise Avenue North, Secaucus NJ	Place record; no time recorded (0)

Types are inferred from content, as the schema does not label them. `gmm_storage_table` (EV-046) holds serialized `CompletedNavigation` and `CommuteSetupNotificationModel` objects whose payloads were not decoded, so **no route is claimed**. `ugc_photos_location_data.db`, `ue3.db` and `inbox_notifications.db` are empty; `offline.db` lists 3 regions. Last Maps sync: 1581277538891 ms = 2020-02-09 19:45:38 UTC.

Media (evidence ID, SHA-256 first 16)	EXIF taken, offset	UTC	GPS	Google Photos row (gphotos0.db, EV-049)
DCIM/Camera/IMG_20200209_132353.jpg (EV-056, 308df52bdab5aa52)	2020-02-09 13:23:53, −05:00 recorded	18:23:53 (GPS clock 18:23:37)	35.660981, −78.879897	local_media 24; utc_timestamp 1581272633139; offset −18000000 ms
DCIM/Camera/IMG_20200213_195030.jpg (EV-057, 2ed43a0a7b01a58f)	2020-02-13 19:50:31, −05:00 recorded	2020-02-14 00:50:31 (GPS 00:50:30)	35.658867, −78.874217	local_media 25; 1581641431590
DCIM/Restored/IMG_20190917_082637.jpg (EV-058, 8c2a100a6a11aecc)	2019-09-17 08:26:37, −04:00 recorded	12:26:37 (GPS 12:26:37)	35.735647, −78.636367	local_media 3; 1568723197683; offset −14400000 ms
DCIM/Camera/2020-02-09-125547759.mp4 (EV-055, 84b124b5e85ed79d)	QuickTime dates all zero; no offset	**Unresolved** (file time 17:55:54)	none	local_media 23; 1581270953000; offset −18000000 ms

All four are Google Pixel 3 files with `is_deleted` 0 and `has_local` 1; the Photos coordinates are the EXIF values rounded to four decimals. The 2019 photo sits in `DCIM/Restored`. The video's Photos time is database-asserted and is not backed by any value in the file. The recorded offsets agree with the GPS clocks to within 16 s and with the creator's UTC-05:00 for February. **Comparison:** Maps place A is 119 m from the 2019 photo, and place B is 148 m from the 2020-02-13 photo (EV-057) and 655 m from the 2020-02-09 photo. **Strength:** neighbourhood-scale agreement from independent databases. **Limits:** a geocoded address is a point on a map and not where anyone stood; B has no time and A predates its photo by 166 days; both records come from one handset; and neither identifies who held the phone, so a visit is not proved. Separately, the 2020-02-09 photo lies 7.7 to 8.4 m from the three WhatsApp location shares sent 18:00 to 18:05 UTC.

9. Timeline, correlations and conclusion (Task 8)

Timeline.csv holds **30 events from 8 artefact families** (Appendix B). Duplicate representations of one message or call (Telephony, Google Messages, Dialer) are counted once. Key events:

#	UTC	Family	Event	Evidence
T06	2020-02-08 20:00:46	TextNow	TextNow "Hi there!" received (first of 9)	EV-034
T10	2020-02-08 20:55:14	WhatsApp	WhatsApp text sent to Josh Hickman	EV-028..030; EV-031..033
T11	2020-02-08 20:57:05	WhatsApp	WhatsApp image received (Osborne computer)	EV-028..030; EV-059
T12	2020-02-08 20:57:49	WhatsApp	WhatsApp "The Osbourne!" sent	EV-028..030
T13	2020-02-08 20:58:12	WhatsApp	WhatsApp meme image sent	EV-028..030; EV-060
T14	2020-02-09 18:00:14	WhatsApp	WhatsApp location share 1 (35.6609133, −78.8798813)	EV-028..030
T16	2020-02-09 18:23:53	Photos/media	Photo taken, GPS 7.7 to 8.4 m from the shares	EV-056; EV-049
T17	2020-02-09 18:41:42	SMS (Telephony)	SMS sent: on my way to the coffee shop	EV-018; EV-023..025
T21	2020-02-14 00:50:31	Photos/media	Photo taken (−05:00), 148 m from Maps place B	EV-057; EV-049; EV-047
T22	2020-02-14 01:29:01	Call log	Outgoing call to Josh Hickman (11 s)	EV-021; EV-026
T24	2020-02-14 01:39:18	Device profile	Profile user 11 created	EV-006; EV-009
T25	2020-02-14 12:50:01	Device profile	User 11 last logged in	EV-006
T26	2020-02-14 13:05:57	Call log	Outgoing call to Josh Hickman (86 s)	EV-021; EV-026
T27	2020-02-14 13:09:03	SMS (Telephony)	SMS sent: messages sync between accounts (WAL-only in Google Messages)	EV-018; EV-023..025
T29	2020-02-14 14:41:18	Call log	Incoming call from Josh Hickman (91 s)	EV-021; EV-026

Correlations. (1) **Identity across five sources:** Josh Hickman / +19195790479 appears in contacts2.db (EV-020), wa.db (EV-031 to EV-033), TextNow contacts (EV-034), the call log (EV-021) and SMS/MMS thread 17 (EV-018). (2) **Message to file:** WhatsApp received an image at 20:57:05, the reply "The Osbourne!" followed 44 s later, and the file in `WhatsApp/Media` shows an Osborne computer and hashes to `media_hash` (EV-059); the sent meme matches row 16 (EV-060). (3) **Location, photo and SMS on 2020-02-09:** WhatsApp location shares 18:00:14 to 18:05:02, a photo with GPS 7.7 to 8.4 m away at 18:23:53, then the SMS "on my way to the coffee shop" at 18:41:42 (WhatsApp, Photos/EXIF, Telephony and Google Messages). (4) **Profiles, calls and messages on 2020-02-14:** a photo at 00:50:31, calls with Josh Hickman at 01:29:01 and 01:30:24, user 11 created at 01:39:18; later user 11 last logged in at 12:50:01, a call at 13:05:57, the SMS "messages sync over between accounts" at 13:09:03 and a reply at 13:10:08. Which profile placed the calls cannot be decided from the records.

Conclusion: unresolved. The records examined show a handset used by two profiles and several accounts, and communications by SMS/MMS, WhatsApp and TextNow with Josh Hickman and with another number. The exchanged content is a short chat, service verification and notification messages, and four images (two vintage computers, a game console and a meme); a location share and a photo place the phone in one area on 2020-02-09. **No examined record contains information that can be attributed to Harbour Operations Ltd, and no record shows an unauthorised disclosure.** The allegation is therefore not supported by what was examined, but it is not contradicted either: coverage is partial and key material is missing (see Section 10), and nothing on the device shows the company's ownership or policy.

Alternative interpretations. (a) **Personal or test use:** the messages themselves say "Have to keep generating data", and the media are generic images, which fits routine activity with no business information. (b) **Another handler:** profile 11 and a second Google account were created on 2020-02-14, and SMS 25 mentions messages syncing "between accounts", so activity that day may belong to a different person. (c) **Content elsewhere or lost:** Signal, Telegram, Viber, Skype, imo and TikTok accounts exist but were not examined, WhatsApp encrypted backups were not opened, and absent rows (Google Messages `_id` 22 to 24, deleted contacts) mean content may be missing. **Additional evidence needed:** the other messaging-app databases; authorised decryption of the WhatsApp backups; Harbour Operations Ltd policy, enrolment or MDM records and a definition of protected information; the holders of +19842032223 and +19102697333 and subscriber records for +19195794674; user 11's encrypted data; and acquisition notes explaining the Magisk artefacts.

10. Limitations

- Only the supplied extraction was examined; applications and accounts beyond those named were not analysed, and no live account was accessed.
- Times: UTC conversions use validated units (milliseconds, or seconds for `pdu.date`) and recorded EXIF offsets. Media times without a recorded offset (the video, zone-less file times) stay **unresolved**; the VM runs on US Eastern time, so only UTC conversions are used.
- Missing or empty records: `data/system_ce/11`, user_de telephony data, the calendar events, the Wi-Fi networks, the Dialer voicemail messages and a WhatsApp image name mismatch (UUID name versus file name) are documented, not explained.
- Codes and decoding: Google Messages protocols 3 and 5 and statuses 100, 207 and 210, `lastLoggedIn` semantics and the Maps record-type labels were not validated against documentation and are reported as observations only.
- Secrets were not read or are redacted (tokens, keys, verification codes, media links). Screenshots are cropped to the terminal window; the first pass of some outputs briefly displayed token strings and was discarded and logged.

11. References

ICDFA, SBT-DF205 supplied Android forensic investigation slides (examination procedures). Hickman, J., Android 10 Pixel 3 dataset, Digital Corpora (URL in Section 2). Android Developers, `android.provider.CallLog.Calls` reference (call type and number presentation constants), developer.android.com/reference/android/provider/CallLog.Calls. Android Open Source Project, `android.content.pm.UserInfo` (user flags FLAG_PRIMARY 0x1, FLAG_ADMIN 0x2, FLAG_INITIALIZED 0x10), android.googlesource.com, frameworks/base.

Appendix A: Supporting screenshots

Screenshots are cropped to the terminal window; raw outputs are in the supporting_evidence folder. Section numbers refer to the report.

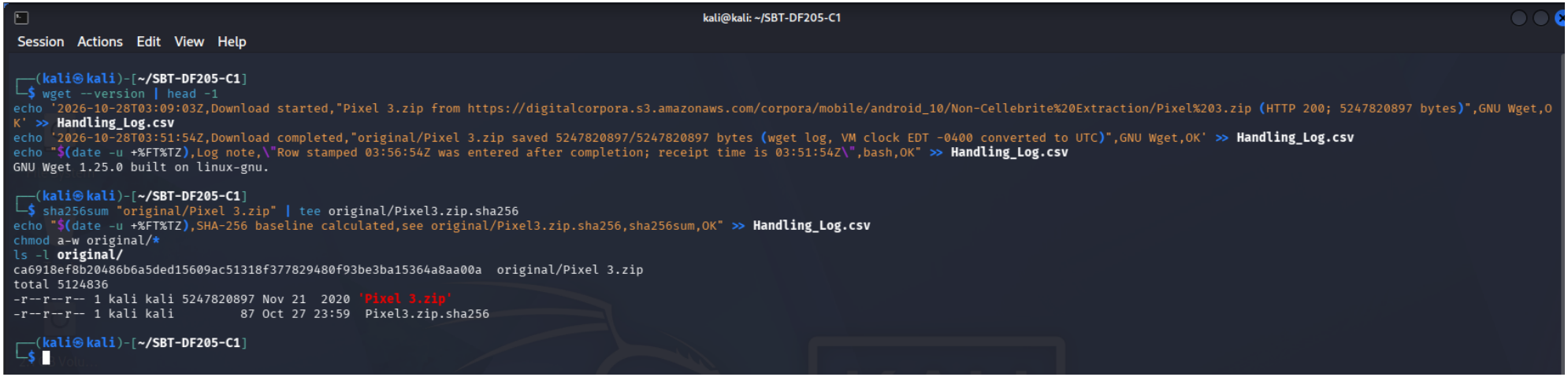


Figure 1: Receipt log rows and baseline SHA-256 of Pixel 3.zip (ca6918ef...a8aa00a); original/ made read-only (Section 2).

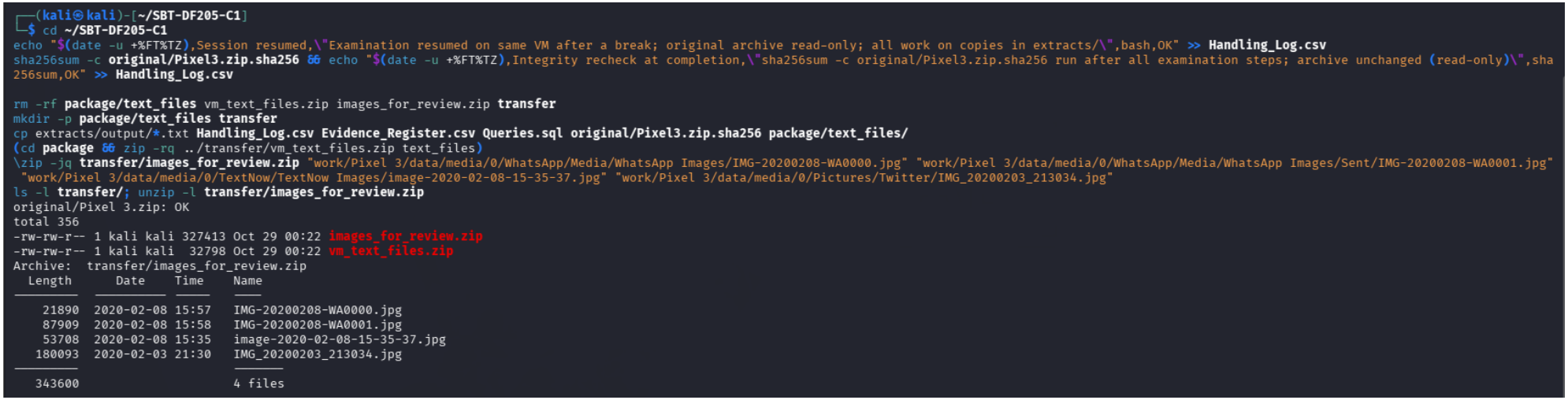


Figure 2: Integrity recheck after all examination steps: original/Pixel 3.zip: OK (Section 2).

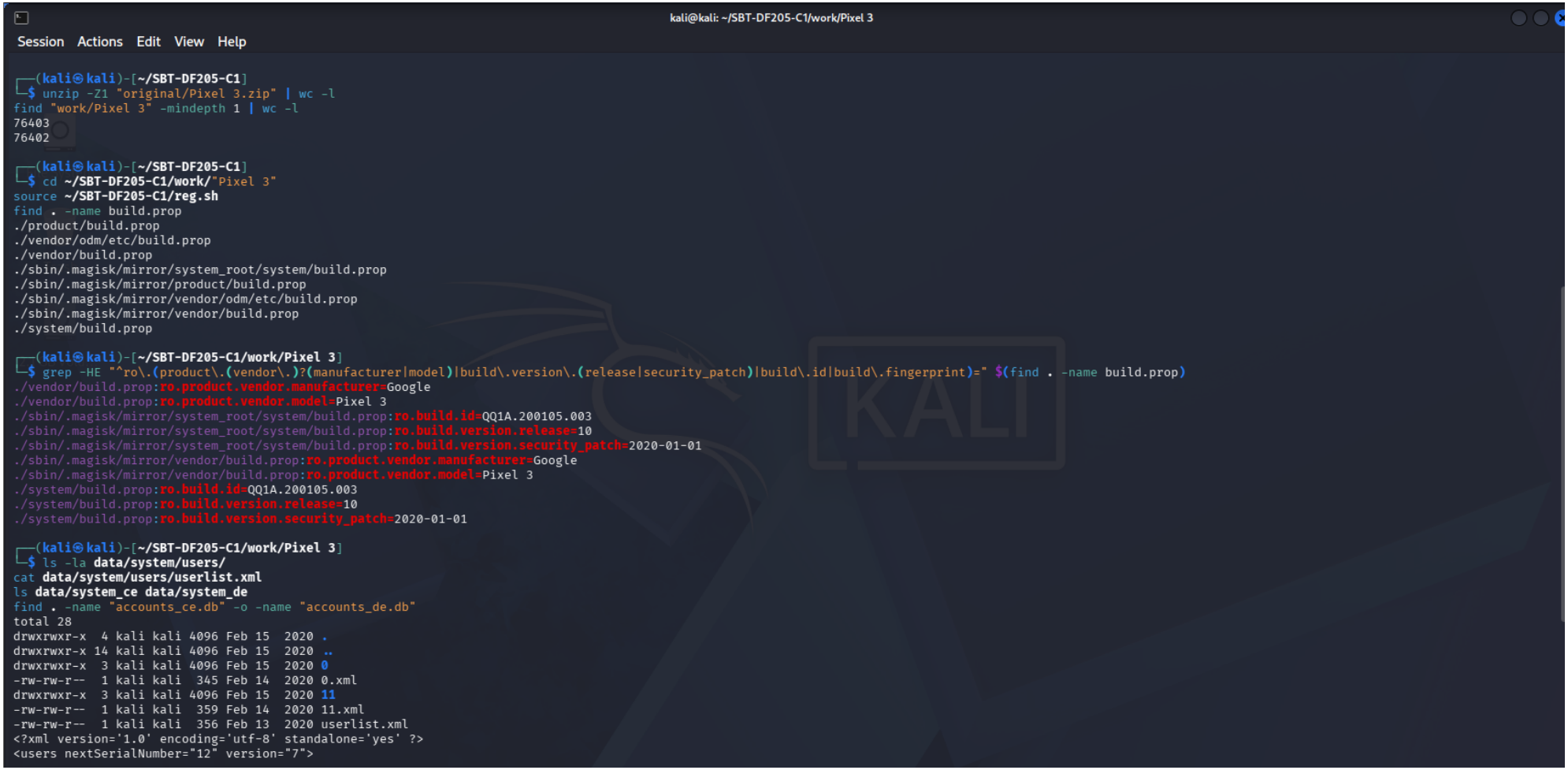


Figure 3: build.prop discovery and property keys: manufacturer, model, release, build ID, security patch (Section 3).

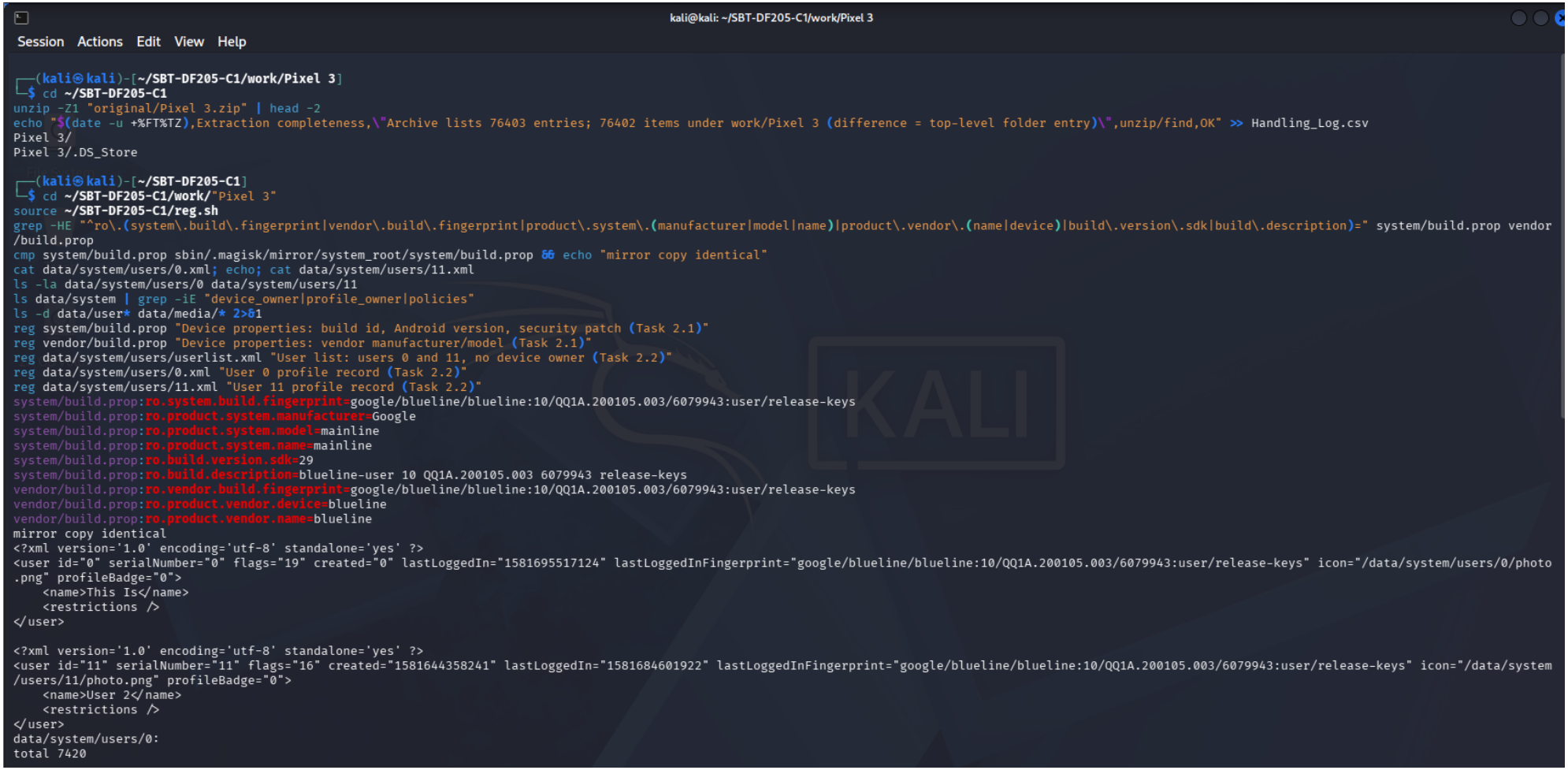


Figure 4: Fingerprint keys, Magisk mirror comparison, user profile XML (users 0 and 11) and register entries EV-002 to EV-006 (Section 3).

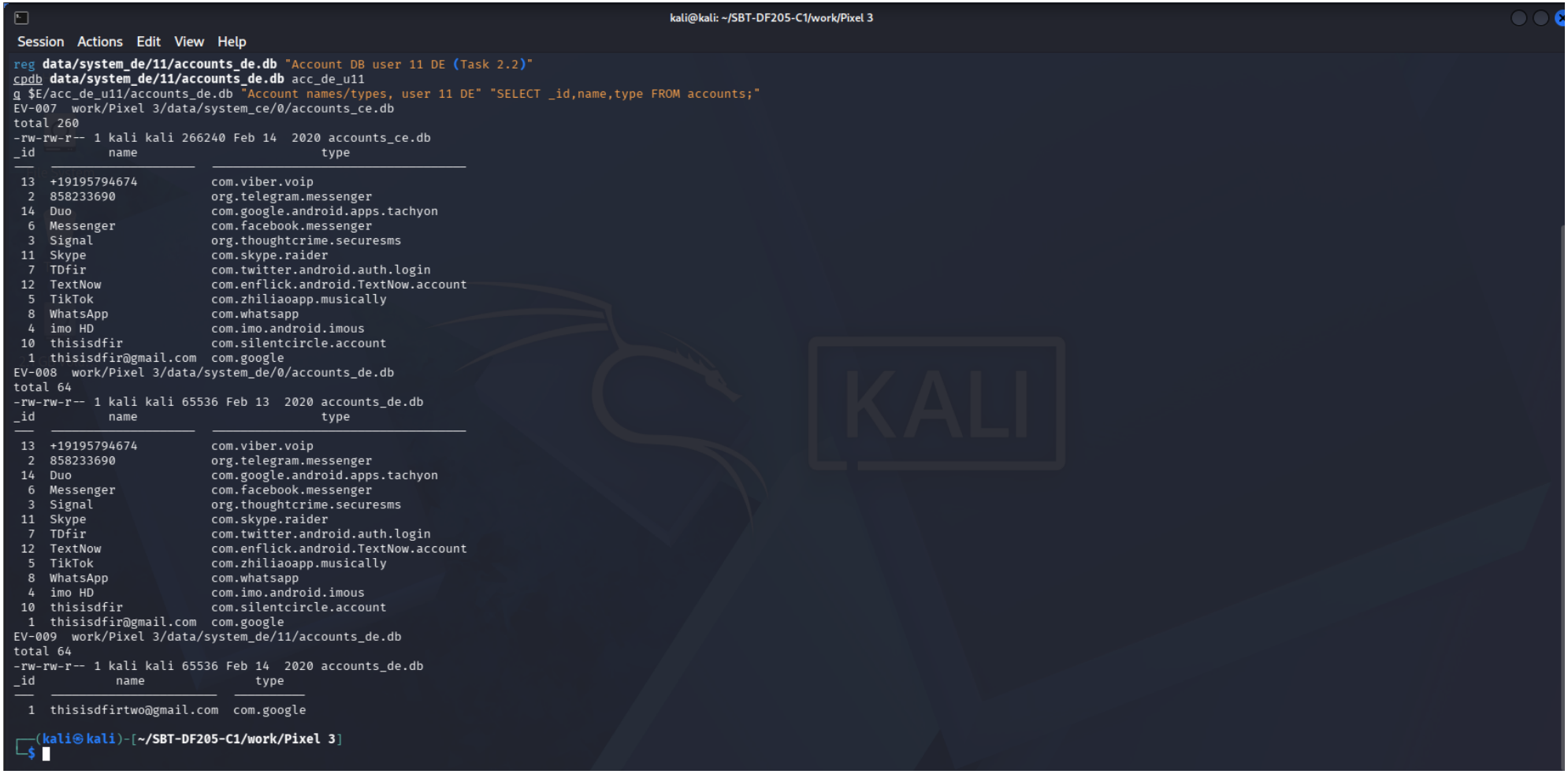


Figure 5: Account names and types from accounts_ce.db and accounts_de.db for users 0 and 11 (Section 3).

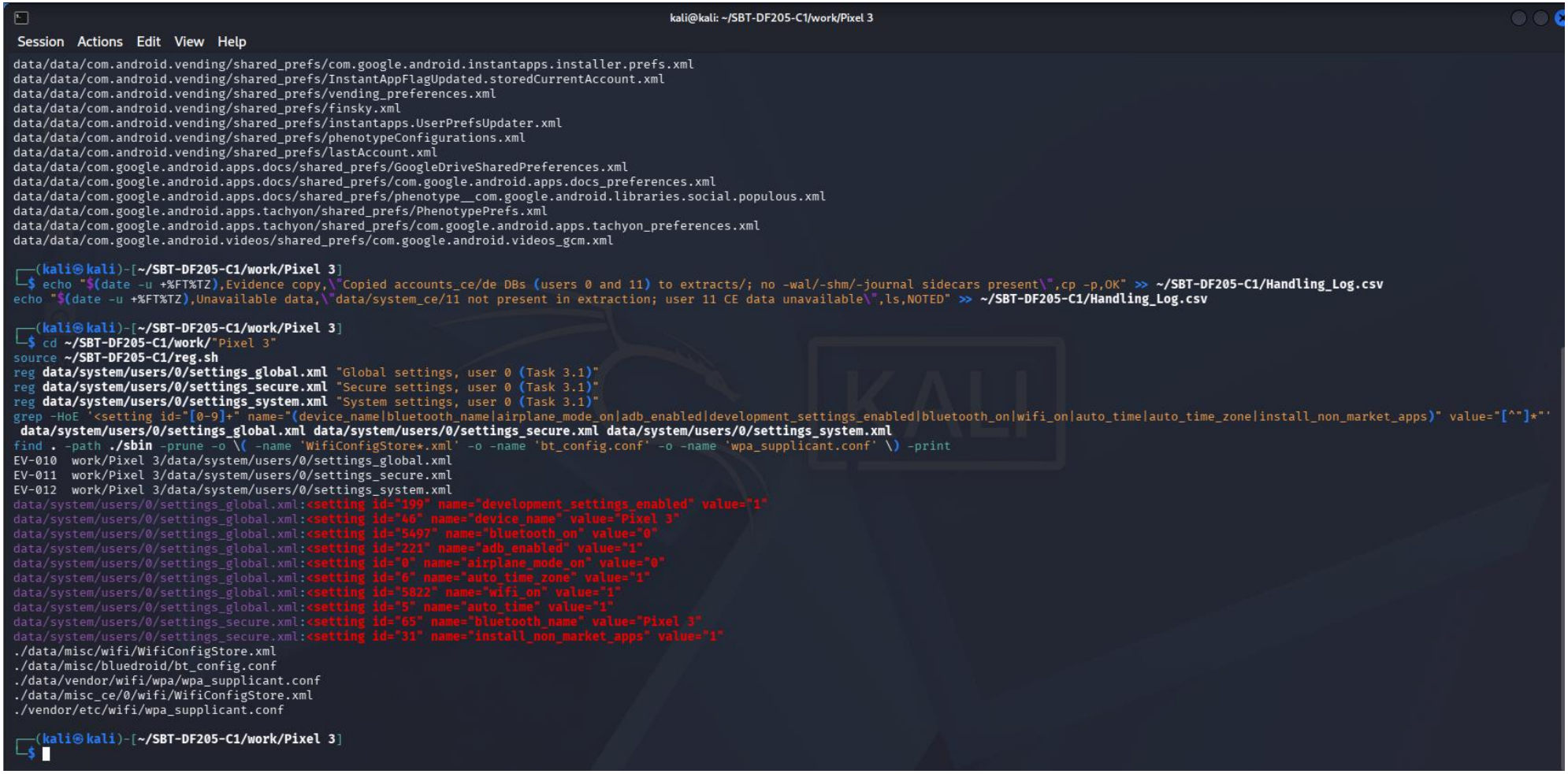


Figure 6: Selected settings keys and values (adb_enabled, development_settings_enabled) and Wi-Fi/Bluetooth file discovery (Section 4).



Figure 7: Thread 17 reconstruction (SMS + MMS, UTC) and five selected SMS with link text redacted (Section 5).

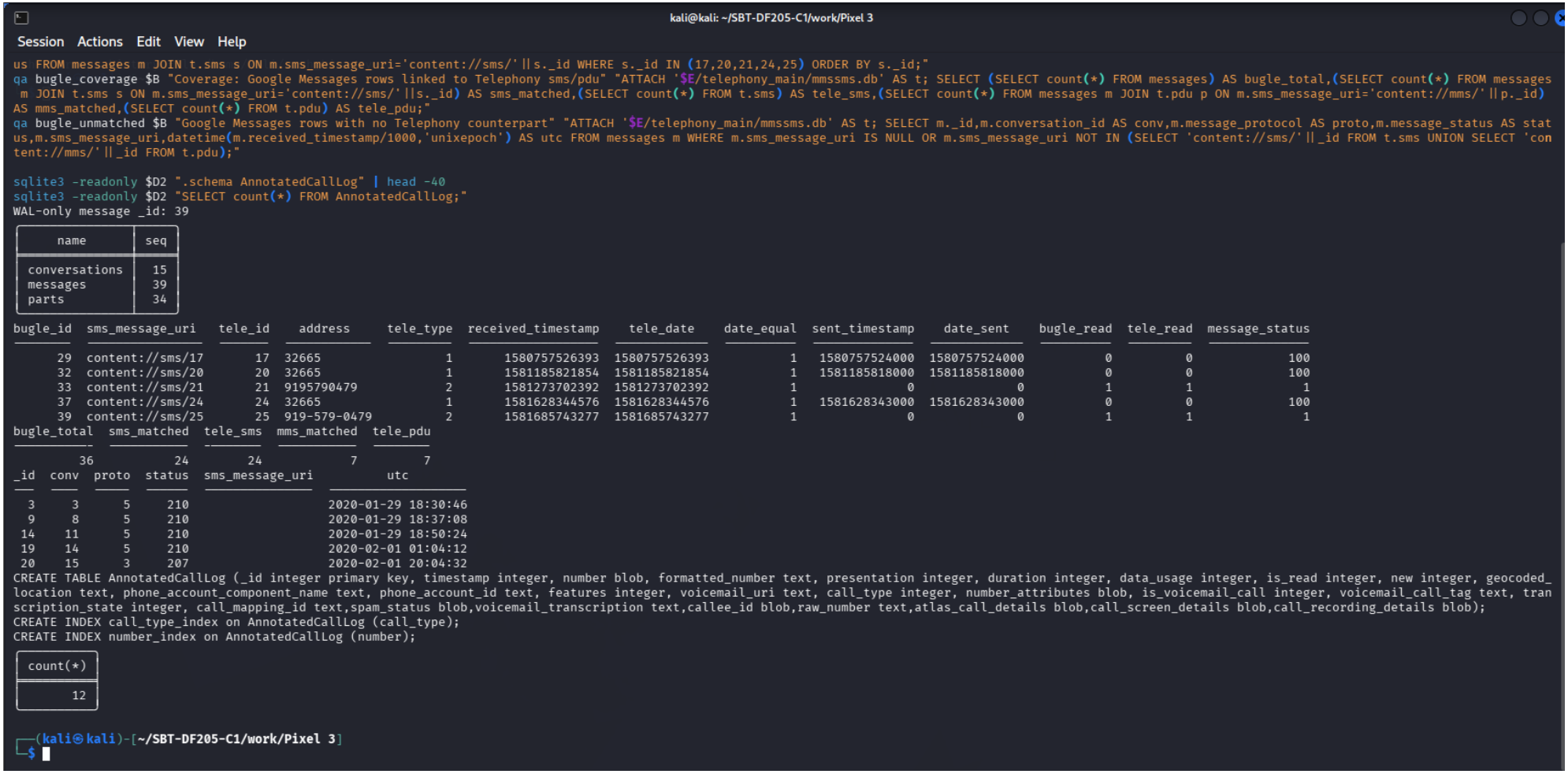


Figure 8: Google Messages WAL-only record (_id 39), Telephony matches, coverage and the five unlinked rows (Section 6).

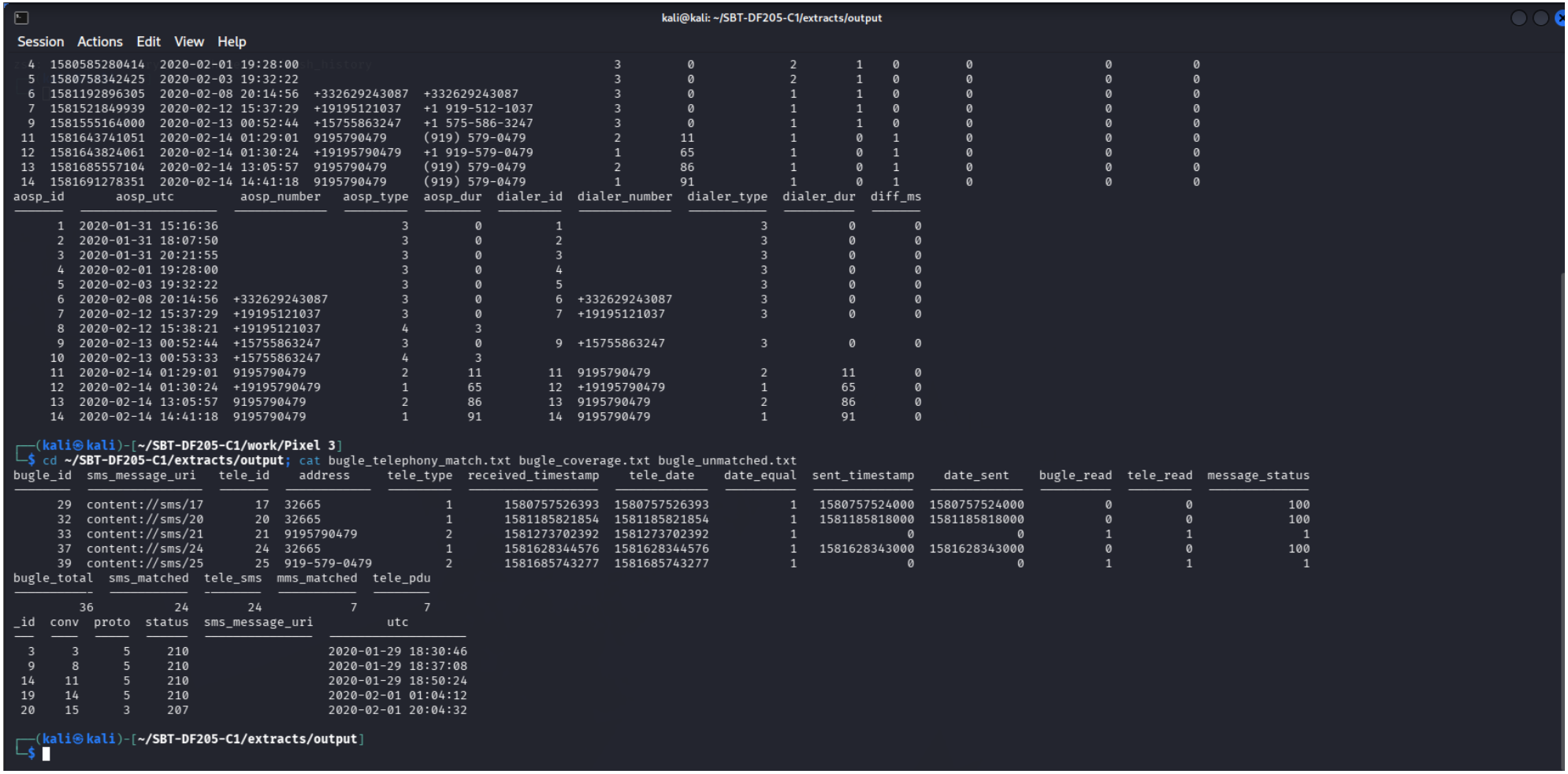


Figure 9: Dialer AnnotatedCallLog versus AOSP call log (12 of 14 rows, 0 ms difference) and the reproducing query (Section 6).

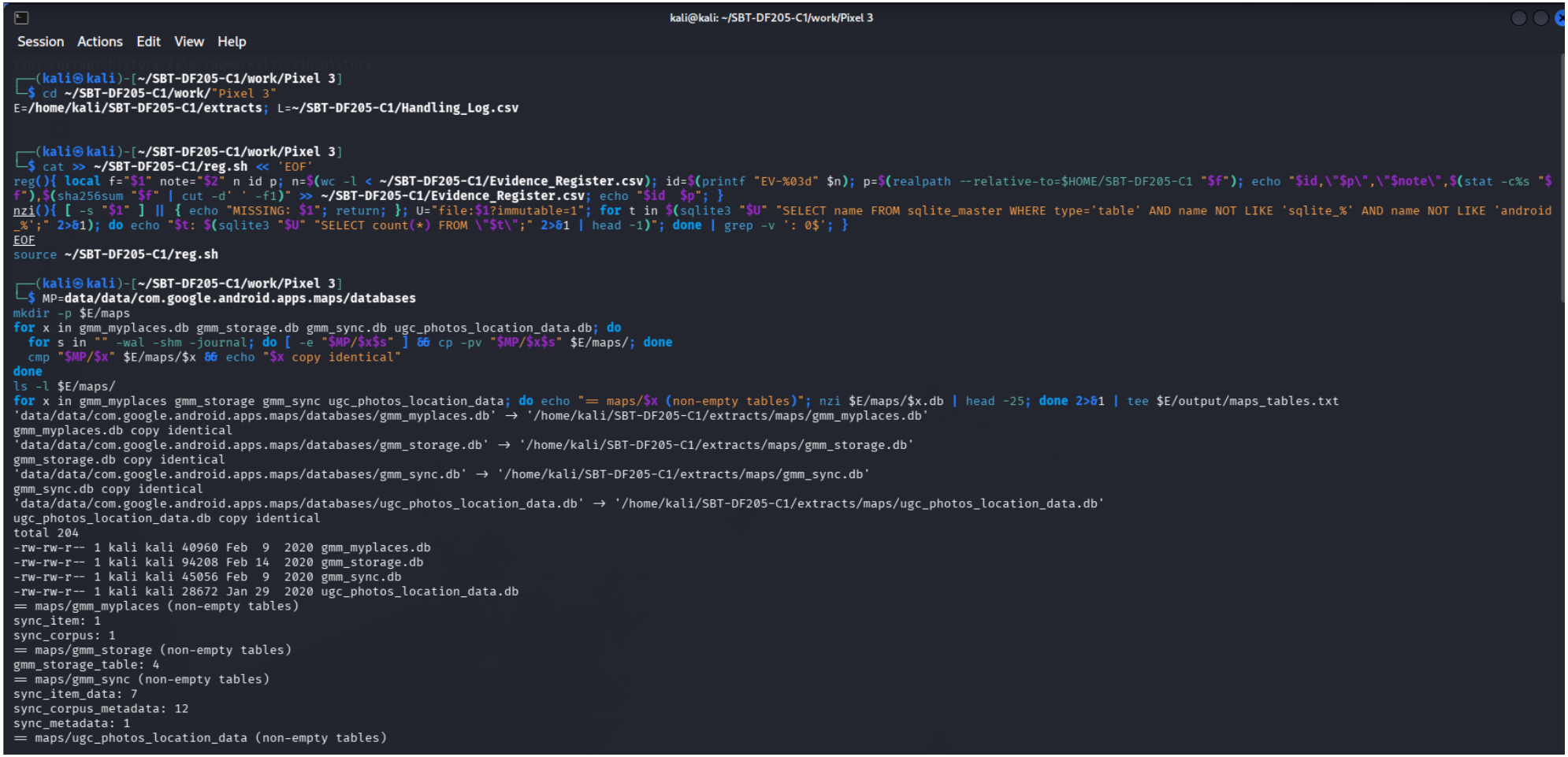


Figure 10: TextNow messages (media links redacted), WhatsApp exchange with receipts, and TextNow contacts (Section 7).

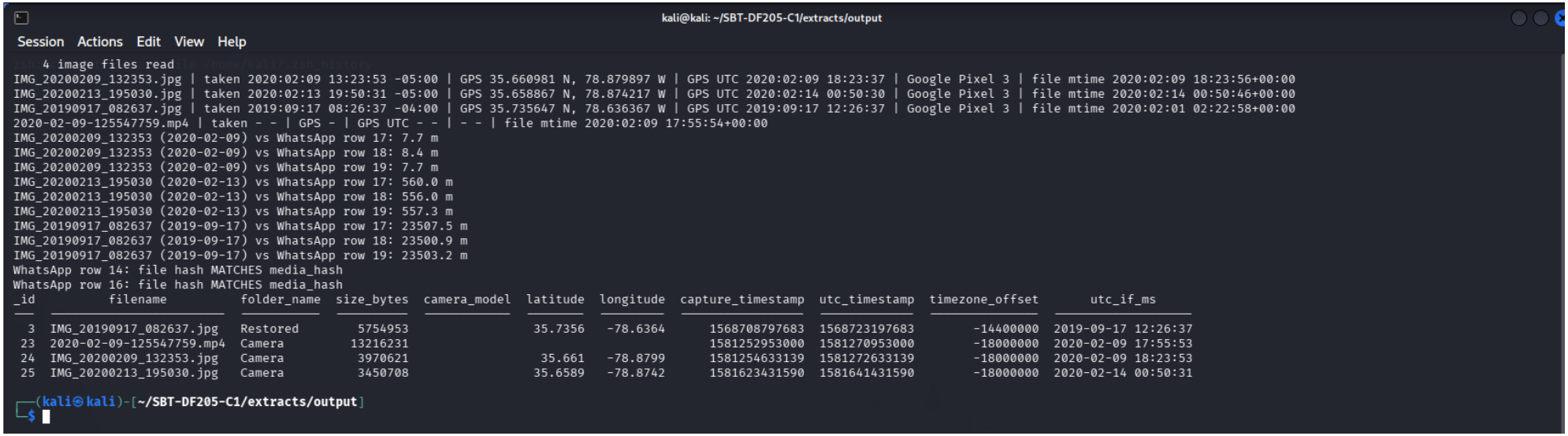


Figure 11: ExifTool summary, WhatsApp file-hash matches, GPS distances and Google Photos rows for the candidate media (Section 8).



Figure 12: Maps place records (sync tokens omitted), epoch conversions and distances to photo GPS (Section 8).

Appendix B: Full timeline (30 events, UTC)

#	UTC	Raw time	Field (unit or format)	Family	Event and note	Evidence
T01	2019-04-04 19:49:03	1554407343604	gmm_myplaces.sync_item.timestamp (ms since 1970-01-01 UTC)	Google Maps	Place record stored for 121 E Tryon Rd, Raleigh NC (35.734602, −78.636654); same feature ID also in gmm_sync.sync_item_data (corpus 11, client_id 1:0) Note: Record creation/sync time not otherwise defined in schema; counted once	EV-045; EV-047
T02	2019-09-17 12:26:37	2019:09:17 08:26:37 −04:00	EXIF DateTimeOriginal + OffsetTimeOriginal (local wall-clock + recorded offset)	Photos/media	Photo IMG_20190917_082637.jpg taken (Pixel 3), GPS 35.735647, −78.636367; file sits in DCIM/Restored Note: GPS clock 12:26:37 UTC agrees; Photos DB utc_timestamp 1568723197683 agrees	EV-058; EV-049
T03	2020-01-29 18:37:08	1580323028883	sms.date (Telephony) (ms since epoch)	SMS (Telephony)	SMS received from short code 47543: WhatsApp verification message (code not reported) Note: Same message also in Google Messages (_id 8); counted once	EV-018; EV-023..025
T04	2020-01-29 18:50:10	1580323810000	messages.timestamp (WhatsApp msgstore) (ms since epoch)	WhatsApp	First 2020 WhatsApp message row: system-type row (type 27) from service number 16505434800 Note: Rows _id 1-7 carry timestamp 0 or 2017-02-14 and appear to be placeholders; excluded	EV-028..030
T05	2020-02-01 20:04:53	1580587493	pdu.date (Telephony) (seconds since epoch)	MMS (Telephony)	Sent MMS 'Here's Signal.' from +19195794674 to 9195790479: first message of thread 17 Note: pdu.date is in seconds (sms.date is ms); Google Messages shows 1580587493369 ms	EV-018
T06	2020-02-08 20:00:46	1581192046000	messages.date (TextNow) (ms since epoch)	TextNow	Incoming TextNow message 'Hi there!' from +19842032223 (first of 9) Note: Direction 1 = incoming (validated: system row Missed call from has direction 1)	EV-034
T07	2020-02-08 20:09:16	1581192556000	messages.date (TextNow) (ms since epoch)	TextNow	Incoming TextNow picture message (message_type 2); attachment TextNow Images/image-2020-02-08-15-35-37.jpg shows a ColecoVision console Note: Media link redacted; time in file name has no offset - unresolved	EV-034; EV-044
T08	2020-02-08 20:10:25	1581192625000	messages.date (TextNow) (ms since epoch)	TextNow	Outgoing TextNow picture message; attachment Pictures/Twitter/IMG_20200203_213034.jpg shows an Atari 800 Note: _id 22 is later than ids 23-24: sort by date, not id	EV-034; EV-043
T09	2020-02-08 20:14:56	1581192896305	calls.date (AOSP) (ms since epoch)	Call log	Missed call from +332629243087 (0 s) Note: Same row in Dialer AnnotatedCallLog (diff 0 ms); counted once	EV-021; EV-026
T10	2020-02-08 20:55:14	1581195314692	messages.timestamp (WhatsApp) (ms since epoch)	WhatsApp	Sent WhatsApp text 'Hi there! I switched over.' to 19195790479 (Josh Hickman); server receipt 20:55:15, read 20:55:18 Note: Counterpart name from wa.db wa_contacts, present only in the WAL	EV-028..030; EV-031..033
T11	2020-02-08 20:57:05	1581195425000	messages.timestamp (WhatsApp) (ms since epoch)	WhatsApp	Received WhatsApp image (image/jpeg, 21,890 B); file IMG-20200208-WA0000.jpg shows an Osborne portable computer Note: File SHA-256 equals media_hash	EV-028..030; EV-059
T12	2020-02-08 20:57:49	1581195469433	messages.timestamp (WhatsApp) (ms since epoch)	WhatsApp	Sent WhatsApp text 'The Osbourne!' 44 s after the image above Note: Content links the reply to the received image	EV-028..030
T13	2020-02-08 20:58:12	1581195492693	messages.timestamp (WhatsApp) (ms since epoch)	WhatsApp	Sent WhatsApp image (87,909 B); file IMG-20200208-WA0001.jpg is a meme captioned: This baby is fully encrypted / Trust me Note: File SHA-256 equals media_hash; media_name is a UUID, file name on disk differs	EV-028..030; EV-060
T14	2020-02-09 18:00:14	1581271214337	messages.timestamp (WhatsApp) (ms since epoch)	WhatsApp	Sent location share (type 16) 35.6609133, −78.8798813 Note: Second share (_id 18) at 18:00:33; a type 5 share (_id 19) at 18:05:02	EV-028..030
T15	2020-02-09 18:05:02	1581271502890	messages.timestamp (WhatsApp) (ms since epoch)	WhatsApp	Sent location share (type 5) 35.6609311, −78.8798382; device receipt 18:07:24, read 18:15:25	EV-028..030
T16	2020-02-09 18:23:53	2020:02:09 13:23:53 −05:00	EXIF DateTimeOriginal + OffsetTimeOriginal (local wall-clock + recorded offset)	Photos/media	Photo IMG_20200209_132353.jpg taken, GPS 35.660981, −78.879897: 7.7-8.4 m from the three WhatsApp location shares Note: GPS clock 18:23:37 UTC; Photos DB utc_timestamp 1581272633139 agrees	EV-056; EV-049

#	UTC	Raw time	Field (unit or format)	Family	Event and note	Evidence
T17	2020-02-09 18:41:42	1581273702392	sms.date (Telephony) (ms since epoch)	SMS (Telephony)	Sent SMS 'I am on my way to the coffee shop. Do you want anything?' to 9195790479 Note: Also Google Messages id 33 (date identical); counted once	EV-018; EV-023..025
T18	2020-02-09 18:43:08	1581273788	pdu.date (Telephony) (seconds since epoch)	MMS (Telephony)	Received MMS 'No, thank you' in thread 17 Note: Google Messages shows 1581273788144 ms	EV-018
T19	2020-02-09 19:45:38	1581277538891	gmm_sync.sync_metadata.last_sync_time (ms since epoch)	Google Maps	Last Maps sync recorded (last_sync_time_for_corpus 1581277538919 agrees) Note: Last-sync time only; no per-record history	EV-047
T20	2020-02-12 15:38:21	1581521901000	calls.date (AOSP) (ms since epoch)	Call log	Voicemail-type call-log row (3 s) for +19195121037, 52 s after a missed call from the same number Note: Present in AOSP calllog.db only; absent from Dialer AnnotatedCallLog (as is _id 10)	EV-021; EV-026
T21	2020-02-14 00:50:31	2020:02:13 19:50:31 −05:00	EXIF DateTimeOriginal + OffsetTimeOriginal (local wall-clock + recorded offset)	Photos/media	Photo IMG_20200213_195030.jpg taken, GPS 35.658867, −78.874217 (148 m from Maps place record M152 Sweet Vista Lane) Note: GPS clock 00:50:30 UTC	EV-057; EV-049; EV-047
T22	2020-02-14 01:29:01	1581643741051	calls.date (AOSP) (ms since epoch)	Call log	Outgoing call to 9195790479 (Josh Hickman), 11 s	EV-021; EV-026
T23	2020-02-14 01:30:24	1581643824061	calls.date (AOSP) (ms since epoch)	Call log	Incoming call from +19195790479, 65 s	EV-021; EV-026
T24	2020-02-14 01:39:18	1581644358241	users/11.xml created (ms since epoch)	Device profile	Secondary profile user 11 ('User 2', flags 16) created; Google account thisisdfirtwo@gmail.com present in accounts_de.db Note: user 11 CE data absent from extraction	EV-006; EV-009
T25	2020-02-14 12:50:01	1581684601922	users/11.xml lastLoggedIn (ms since epoch)	Device profile	Profile user 11 last logged in Note: Meaning of lastLoggedIn not documented in the supplied material	EV-006
T26	2020-02-14 13:05:57	1581685557104	calls.date (AOSP) (ms since epoch)	Call log	Outgoing call to 9195790479, 86 s	EV-021; EV-026
T27	2020-02-14 13:09:03	1581685743277	sms.date (Telephony) (ms since epoch)	SMS (Telephony)	Sent SMS 'Wow, the messages sync over between accounts.'; exists in bugle_db only through the WAL (messages._id 39) Note: Absent from a read of bugle_db without its -wal file	EV-018; EV-023..025
T28	2020-02-14 13:10:08	1581685808	pdu.date (Telephony) (seconds since epoch)	MMS (Telephony)	Received MMS 'That is interesting.' 65 s after the sent SMS	EV-018
T29	2020-02-14 14:41:18	1581691278351	calls.date (AOSP) (ms since epoch)	Call log	Incoming call from 9195790479, 91 s	EV-021; EV-026
T30	2020-02-14 15:51:57	1581695517124	users/0.xml lastLoggedIn (ms since epoch)	Device profile	Profile user 0 ('This Is', flags 19) last logged in Note: created=0 in the file: no creation time recorded	EV-005